MAT1100 Homework 2

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Question 4

(a) *Proof.* $|G| = 56 = 7 \cdot 8$ hence by Sylow's theorem

$$n_7 = 1 \mod 7 \quad n_7 \mid 8$$

hence $n_7 = 1$ or 8 and

$$n_8 = 1 \mod 2 \quad n_8 \mid 7$$

hence $n_8 = 1$ or 7.

Now if $n_7 = 1$ then the unique Sylow 8-subgroup of G must be normal similarly if $n_8 = 1$ Hence it suffices to show that it is not possible for $n_7 = 8$ and $n_8 = 1$. To prove this, suppose it holds for contradiction. Groups of order 7 are cyclic and any non-identity element generates the group. Hence if P, Q are Sylow 7-subgroups they are either equal or intersect trivially. Thus the 8 Sylow 7-subgroups of G contribute $6 \cdot 8 = 48$ distinct elements of order 7.

Now if $H_1, \ldots, H_7$ are the seven distinct Sylow 8-subgroups of G. Pick a non-identity element x_1 in H_1 then since H_2 has order 8 there exists a non-identity element $x_2 \in H_2$ not equal to x_1 and iterating this process we find non-identity $x_i \in H_i$ such that $x_1, \ldots x_7$ are distinct. Now since not all of the H_i are equal $|\bigcup_i H_i| > 8$ hence there exists a non-identity x_8 in some H_i not equal to any of the x_i.

These are 8 even order elements in G which are distinct from the 48 elements of order 7 above. However adding the identity this implies that $|G| \geq 1 + 8 + 48 = 57$ which contradicts the fact that G has order 56.

Hence G must have a normal subgroup. □

(b) *Proof.* Suppose G has order $2^k p$ for p an odd prime and k a natural number such that $2^k p > p!$. Since $2 \nmid p$ Sylow's theorem guarantees the existence of $H \leq G$ such that $|H| = 2^k$ and hence $[G : H] = p$.

Now let G act on the set of cosets of H in G by left multiplication. Suppose $\pi : G \to S_{G/H} = S_p$ is the permutation representation of this action. First since $p > 1$, $G \setminus H \neq \varnothing$ and hence if $g \in G \setminus H$, $gH \neq H$ and so this action is not trivial, i.e. π is not the zero map and hence $\ker \pi \neq G$.

Now to show that G cannot be simple it suffices to show that π is not injective as then $\ker \pi$ is a non-trivial proper normal subgroup of G. This follows as $|S_p| = p!$ and $|G| = 2^k p > p!$ and if π were injective this would imply that

$$|G| \leq |S_p|$$

□